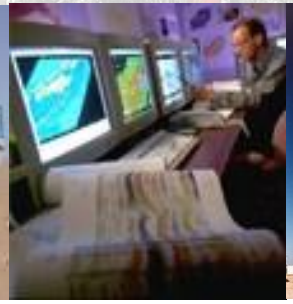
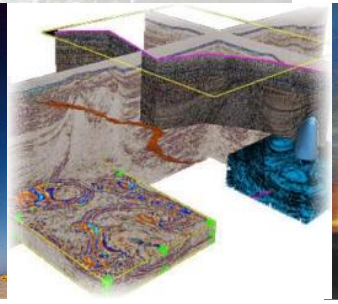
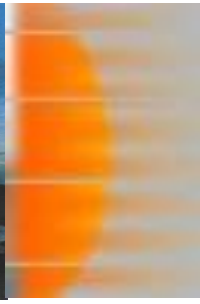
ENG-3020
RENEWABLE ENERGY SOURCES





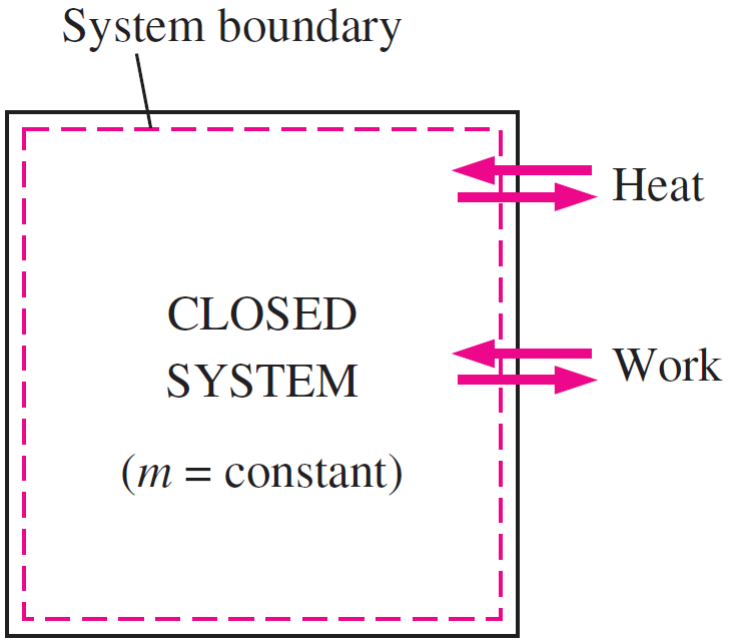




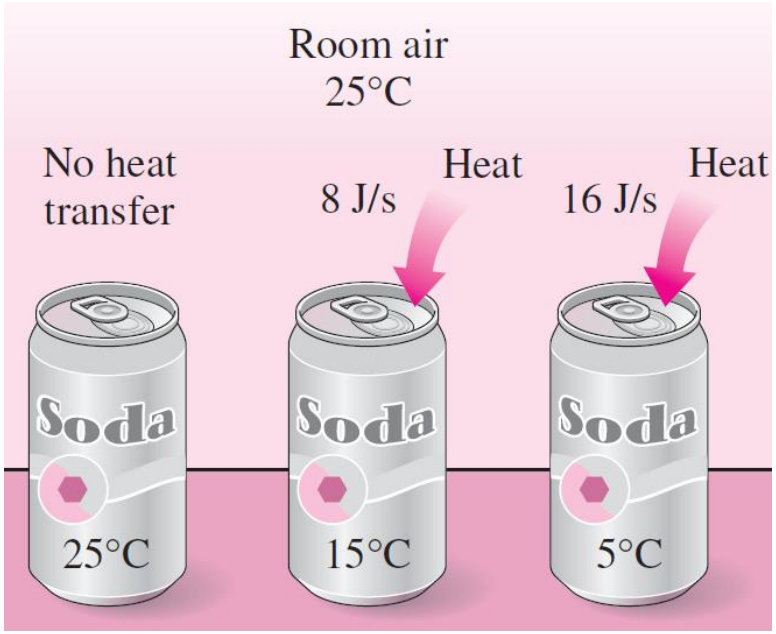
ENERGY

Energy transfer by heat

Heat is defined as *the form of energy that is transferred between two systems (or a system and its surroundings) by virtue of a temperature difference*



Energy can cross the boundaries of a closed system in the form of heat and work.

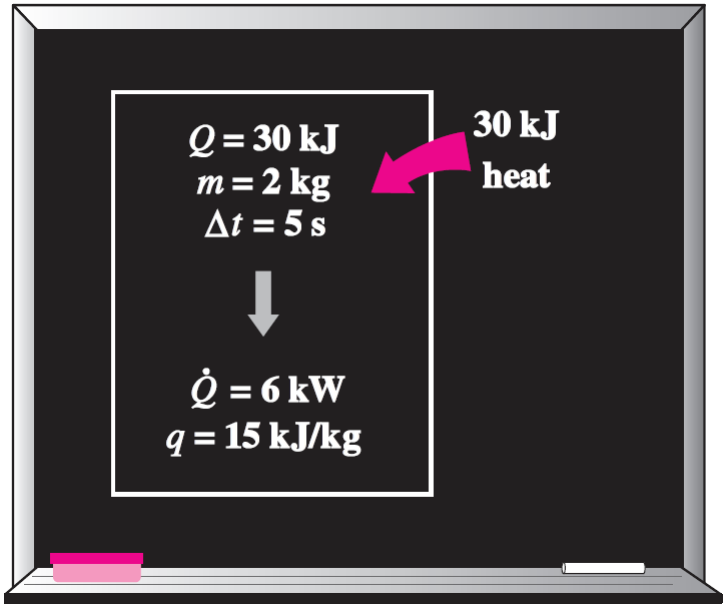


Temperature difference is the driving force for heat transfer. The larger the temperature difference, the higher is the rate of heat transfer.

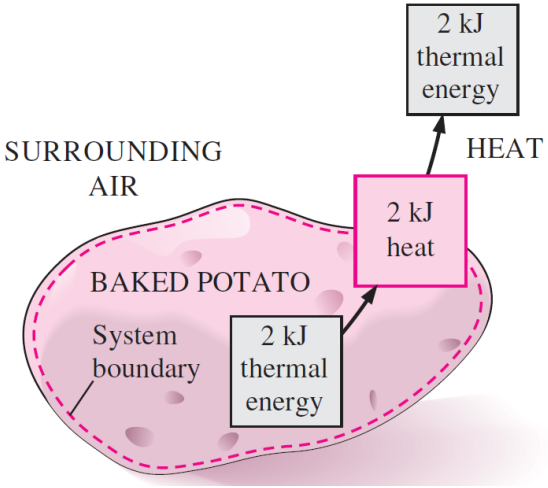
Heat transfer

Heat transfer *per unit mass*

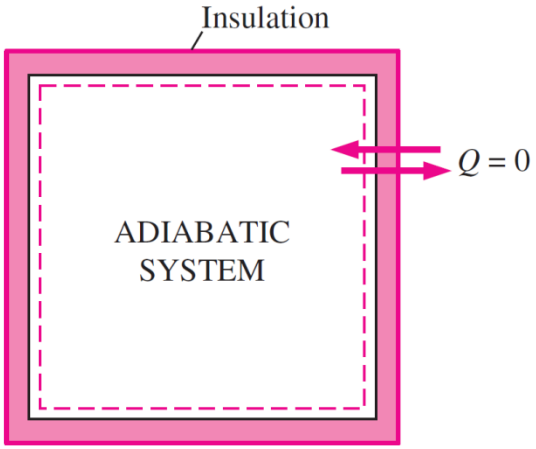
$$q = \frac{Q}{m} \qquad (\text{kJ/kg})$$



the *rate of heat transfer* ($\dot{Q}$) (kW) (the amount of heat transferred per unit time)



Energy is recognized as heat transfer only as it crosses the system boundary



During an adiabatic process, a system exchanges no heat with its surroundings.

Energy Transfer

Energy can be transferred to or from a given mass by two mechanisms:

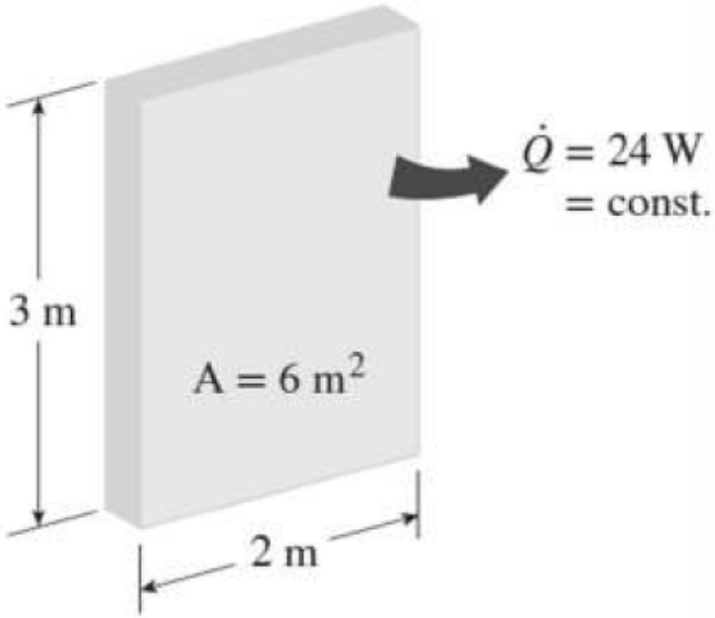
- heat transfer

- work

Heat transfer rate: The amount of heat transferred per unit time.

Heat flux: The rate of heat transfer per unit area normal to the direction of heat transfer.

Power: The work done *per unit time.*



$$\dot{q} = \frac{\dot{Q}}{A} = \frac{24\ \text{W}}{6\ \text{m}^2} = 4\ \text{W/m}^2$$

Heat flux is heat transfer *per unit time* and *per unit area*, and is equal to $\dot{q} = \dot{Q}/A$ when $\dot{Q}$ is uniform over the area A.

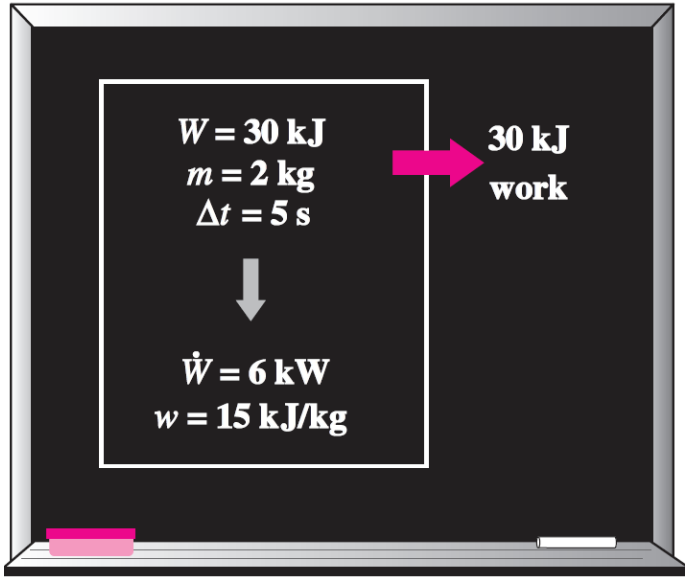
$$Q = \int_0^{\Delta t} \dot{Q}\, dt$$

$$Q = \dot{Q}\,\Delta t \qquad \text{(kJ)}$$

$$\dot{q} = \frac{\dot{Q}}{A} \qquad \text{(W/m}^2\text{)}$$

Energy transfer by work

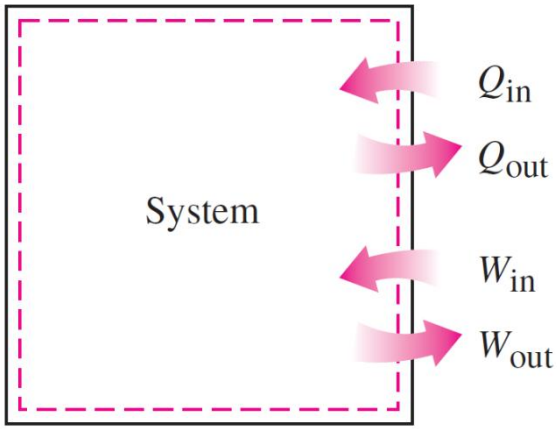
- Work, like heat, is an energy interaction between a system and its surroundings.
- More specifically, *work is the energy transfer associated with a force acting through a distance*. A rising piston, a rotating shaft, and an electric wire crossing the system boundaries are all associated with work interactions.



The work done *per unit time* is called **power** and is denoted by $\dot{W}$

The work done *per unit mass*

$$w = \frac{W}{m} \qquad (\text{kJ/kg})$$



Specifying the directions of heat and work.

MECHANICAL FORM OF WORK

There are two requirements for a work interaction between a system and its surroundings to exist:

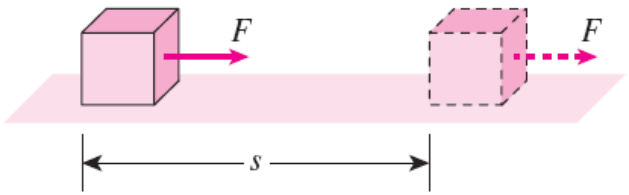
- there must be a ***force*** acting on the boundary.
- the boundary must ***move***.

Work = Force × Distance

$$W = Fs \qquad (\text{kJ})$$

When force is not constant

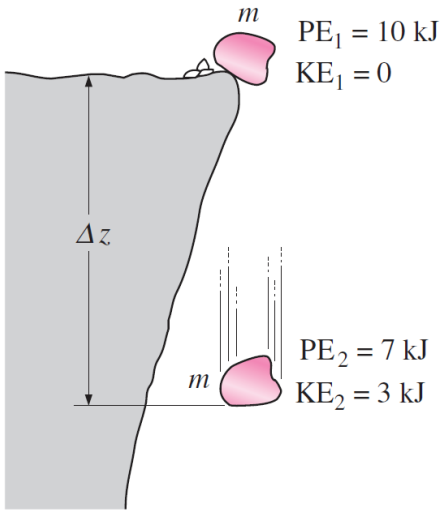
$$W = \int_1^2 F\, ds \qquad (\text{kJ})$$



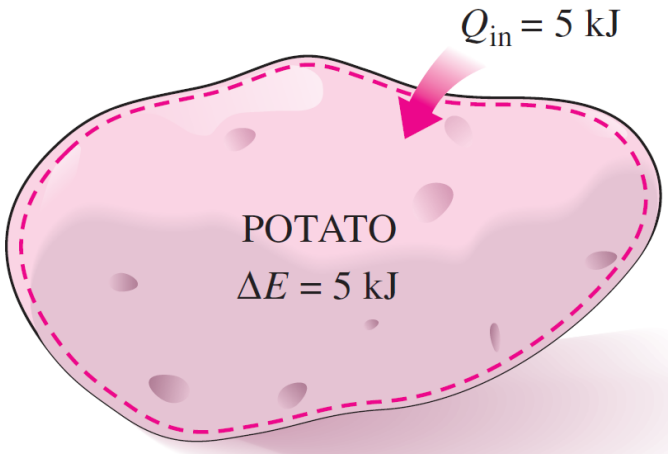
The work done is proportional to the force applied (F) and the distance traveled (s).

The First Law of Thermodynamics

- Based on experimental observations, the first law of thermodynamics states that ***energy can be neither created nor destroyed during a process; it can only change forms***.
- Also known as *the conservation of energy principle*



Energy cannot be created or destroyed; it can only change forms



The increase in the energy of a potato in an oven is equal to the amount of heat transferred to it.

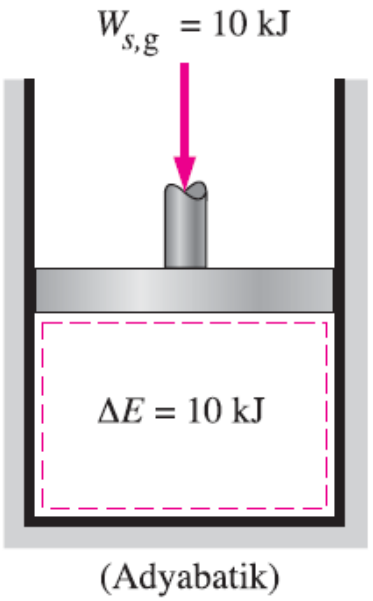
Energy Balance

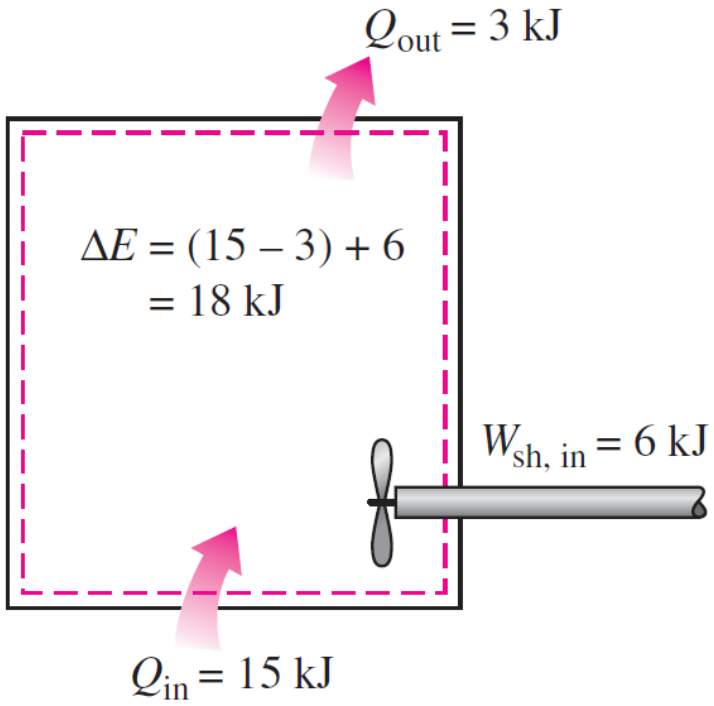
$$\Delta E = \Delta U + \Delta \text{KE} + \Delta \text{PE}$$

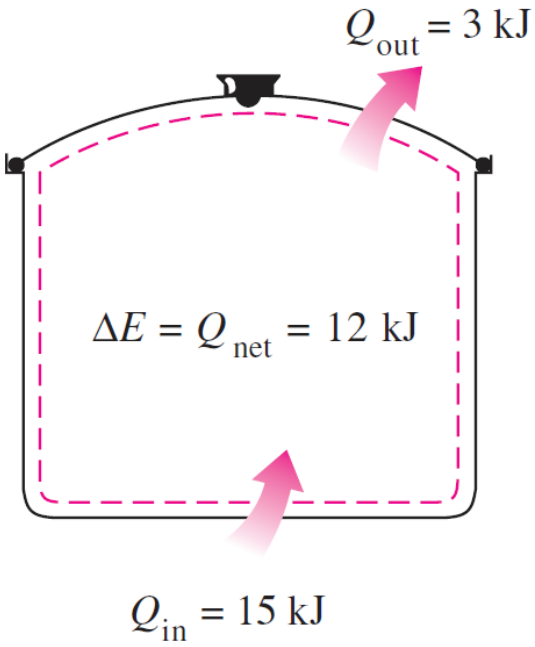
$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

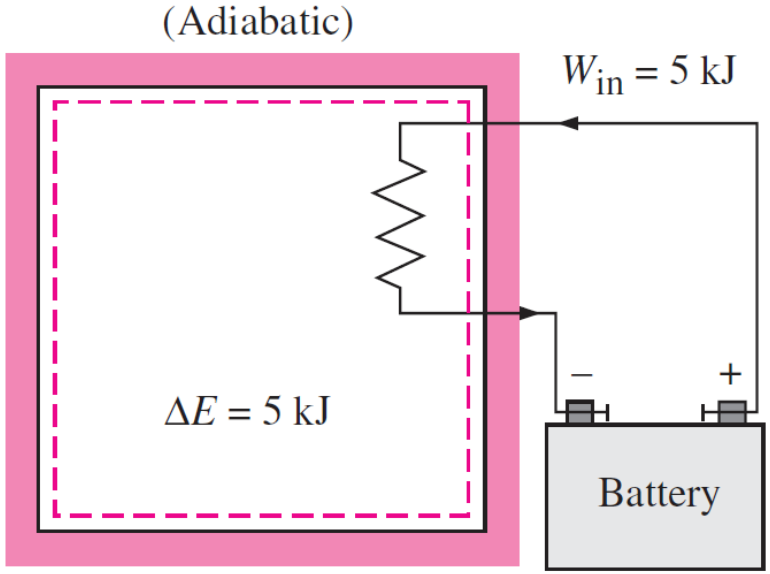
$$(Q_{\text{in}} - Q_{\text{out}}) + (W_{\text{in}} - W_{\text{out}}) + (E_{\text{mass,in}} - E_{\text{mass,out}})$$

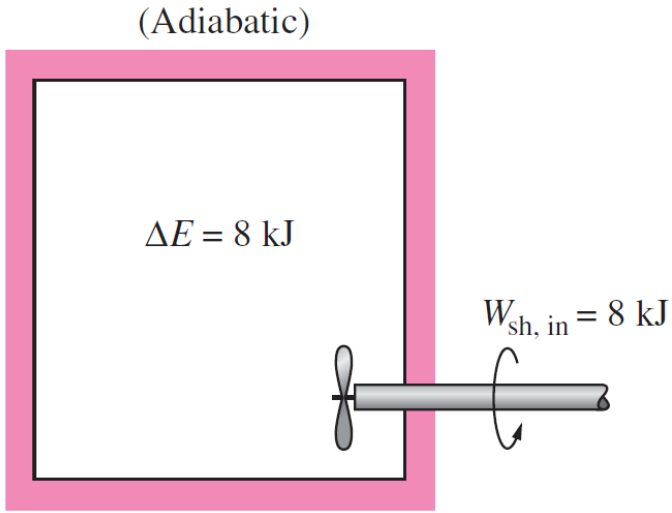
Heat Transfer Work Transfer Mass Flow

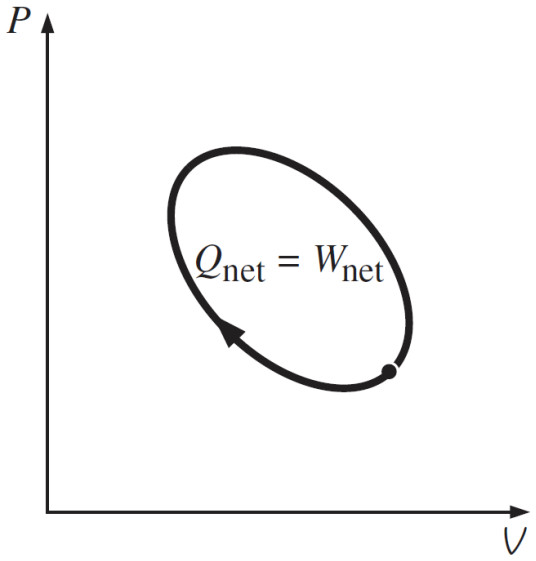





$Q_{out} = 3$ kJ
$\Delta E = Q_{net} = 12$ kJ
$Q_{in} = 15$ kJ

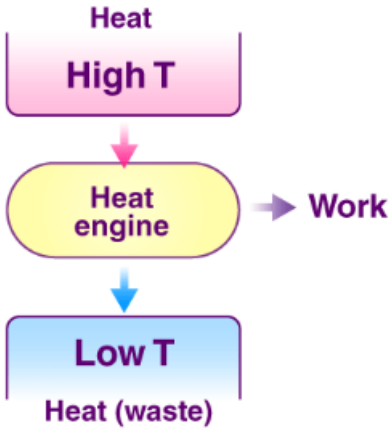

(Adiabatic)
$W_{in} = 5$ kJ
$\Delta E = 5$ kJ
–
+
Battery


(Adiabatic)
$\Delta E = 8$ kJ
$W_{sh, in} = 8$ kJ



For a closed system undergoing a **cycle**, the initial and final states are identical, and thus $\Delta E_{system} = E_2 - E_1 = 0$. Thus $Q_{net} = W_{net}$

$$E_{in} - E_{out} = (Q_{in} - Q_{out}) + (W_{in} - W_{out}) + (E_{mass,in} - E_{mass,out}) = \Delta E_{system}$$

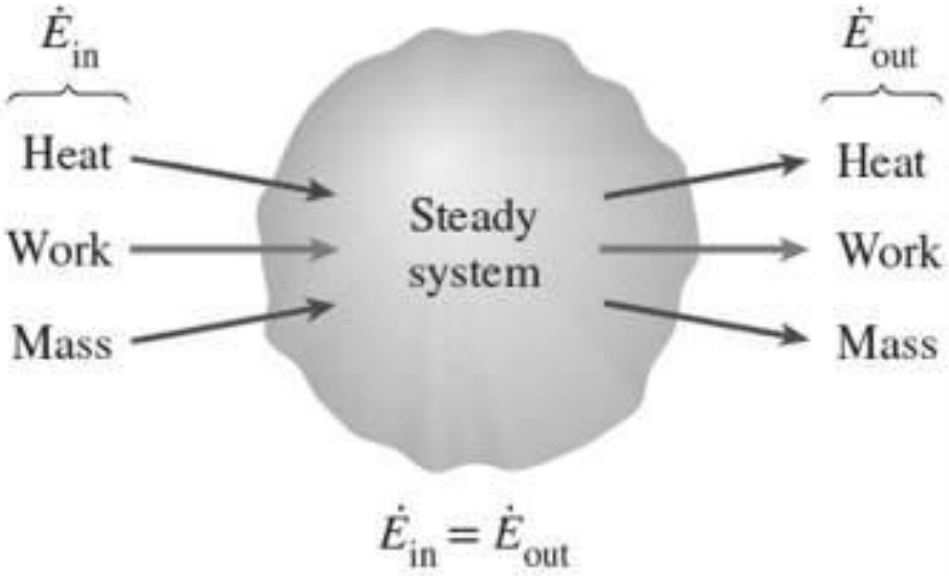


Energy Balance

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

$$E_{in} - E_{out} = dE_{system}/dt$$

$$E_{in} = E_{out} \quad \text{(kW)}$$ **Steady-flow process**



In steady operation, the rate of energy transfer to a system is equal to the rate of energy transfer from the system.

Energy Balance for Closed Systems



A closed system consists of a fixed mass.

The total energy E for most systems encountered in practice consists of the internal energy U.

This is especially the case for stationary systems since they don't involve any changes in their velocity or elevation during a process.

Stationary closed system:

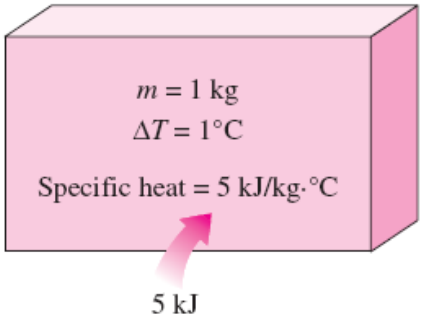
$$E_{in} - E_{out} = \Delta U = mc_v \Delta T \qquad \text{(kJ)}$$

Stationary closed system, no work:

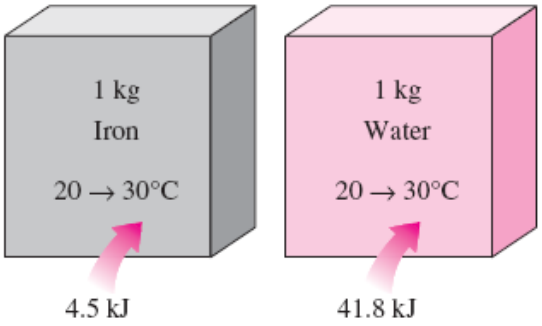
$$Q = mc_v \Delta T \qquad \text{(kJ)}$$

In the absence of any work interactions, the change in the energy content of a closed system is equal to the net heat transfer.

SPECIFIC HEATS



Specific heat is the energy required to raise the temperature of a unit mass of a substance by one degree in a specified way.

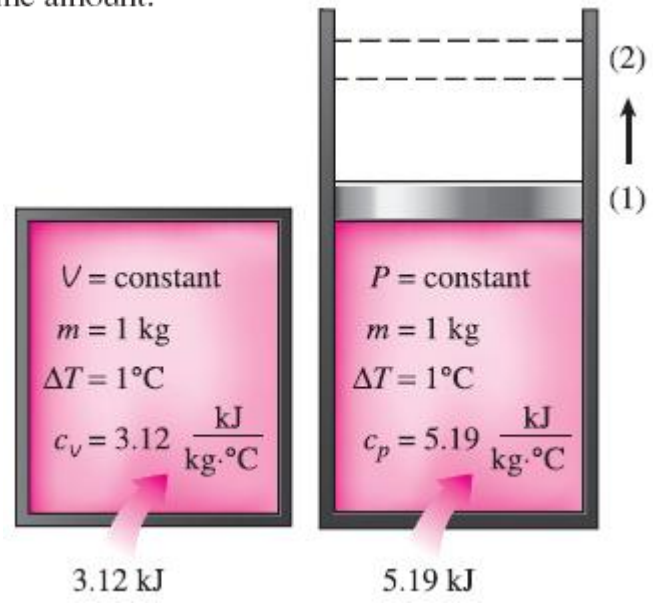


It takes different amounts of energy to raise the temperature of different substances by the same amount.

Specific heat at constant volume, c_v: The energy required to raise the temperature of the unit mass of a substance by one degree as the volume is maintained constant.

Specific heat at constant pressure, c_p: The energy required to raise the temperature of the unit mass of a substance by one degree as the pressure is maintained constant.

Constant-volume and constant-pressure specific heats c_v and c_p (values are for helium gas).

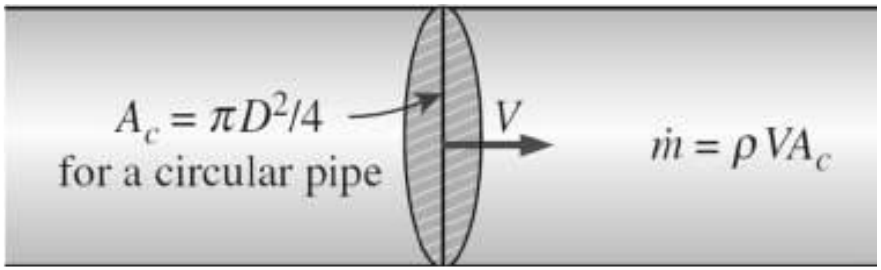


Energy Balance for Steady-Flow Systems

Control volume: A large number of engineering devices such as water heaters and car radiators involve mass flow in and out of a system.

Most control volumes are analyzed under steady operating conditions.

Steady: No change with time at a specified location.



The mass flow rate of a fluid at a cross section is equal to the product of the fluid density, average fluid velocity, and the cross-sectional area.

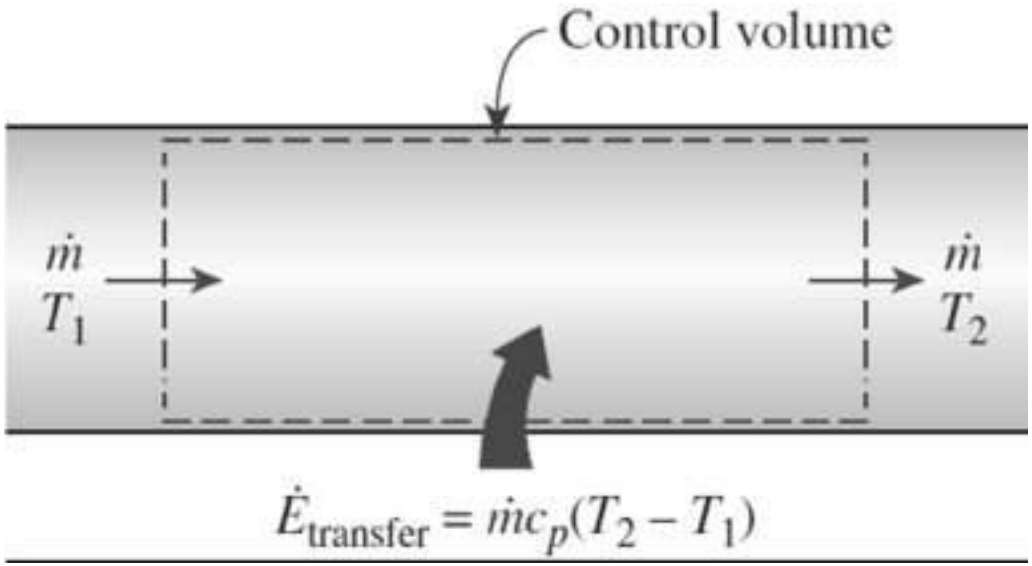
Mass flow rate: The amount of mass flowing through a cross section of a flow device per unit time

$$\dot{m} = \rho V A_c \qquad (\text{kg/s})$$

Volume flow rate: The volume of a fluid flowing through a pipe or duct per unit time.

$$\dot{V} = V A_c = \frac{\dot{m}}{\rho} \qquad (\text{m}^3/\text{s})$$

$$\dot{Q} = \dot{m}\Delta h = \dot{m} c_p \Delta T \qquad (\text{kW})$$



Under steady conditions, the net rate of energy transfer to a fluid in a control volume is equal to the rate of increase in the energy of the fluid stream flowing through the control volume.

Saturation Temperature and Saturation Pressure

The temperature at which water starts boiling depends on the pressure; therefore, if the pressure is fixed, so is the boiling temperature.
Water boils at 100 °C at 1 atm pressure.

Saturation temperature (T_{sat}): The temperature at which a pure substance changes phase at a given pressure.

Saturation pressure (P_{sat}): The pressure at which a pure substance changes phase at a given temperature.

Saturation (Boiling) Pressure of Water at Various Temperatures

Temperature T, °C	Saturation Pressure P_{sat}, kPa
−10	0.260
−5	0.403
0	0.611
5	0.872
10	1.23
15	1.71
20	2.34
25	3.17
30	4.25
40	7.38
50	12.35
100	101.3 (1 atm)
150	475.8
200	1554
250	3973
300	8581

Latent heat: The amount of energy absorbed or released during a phase-change process.

Latent heat of fusion: The amount of energy absorbed during melting. It is equivalent to the amount of energy released during freezing.

Latent heat of vaporization: The amount of energy absorbed during vaporization and it is equivalent to the energy released during condensation.

The magnitudes of the latent heats depend on the temperature or pressure at which the phase change occurs.

At 1 atm pressure, the latent heat of fusion of water is 334 kJ/kg and the latent heat of vaporization is 2257 kJ/kg.

The atmospheric pressure, and thus the boiling temperature of water, decreases with elevation.

Variation of the standard atmospheric pressure and the boiling (saturation) temperature of water with altitude

Elevation, m	Atmospheric pressure, kPa	Boiling temperature, °C
0	101.33	100.0
1,000	89.55	96.5
2,000	79.50	93.3
5,000	54.05	83.3
10,000	26.50	66.3
20,000	5.53	34.7

HEAT TRANSFER

HEAT TRANSFER

Heat: The form of energy that can be transferred from one system to another as a result of temperature difference.

Thermodynamics: A thermodynamic analysis is concerned with the amount of heat transfer as a system undergoes a process from one equilibrium state to another.

Heat transfer: The science that deals with the determination of the rates of such energy transfers.

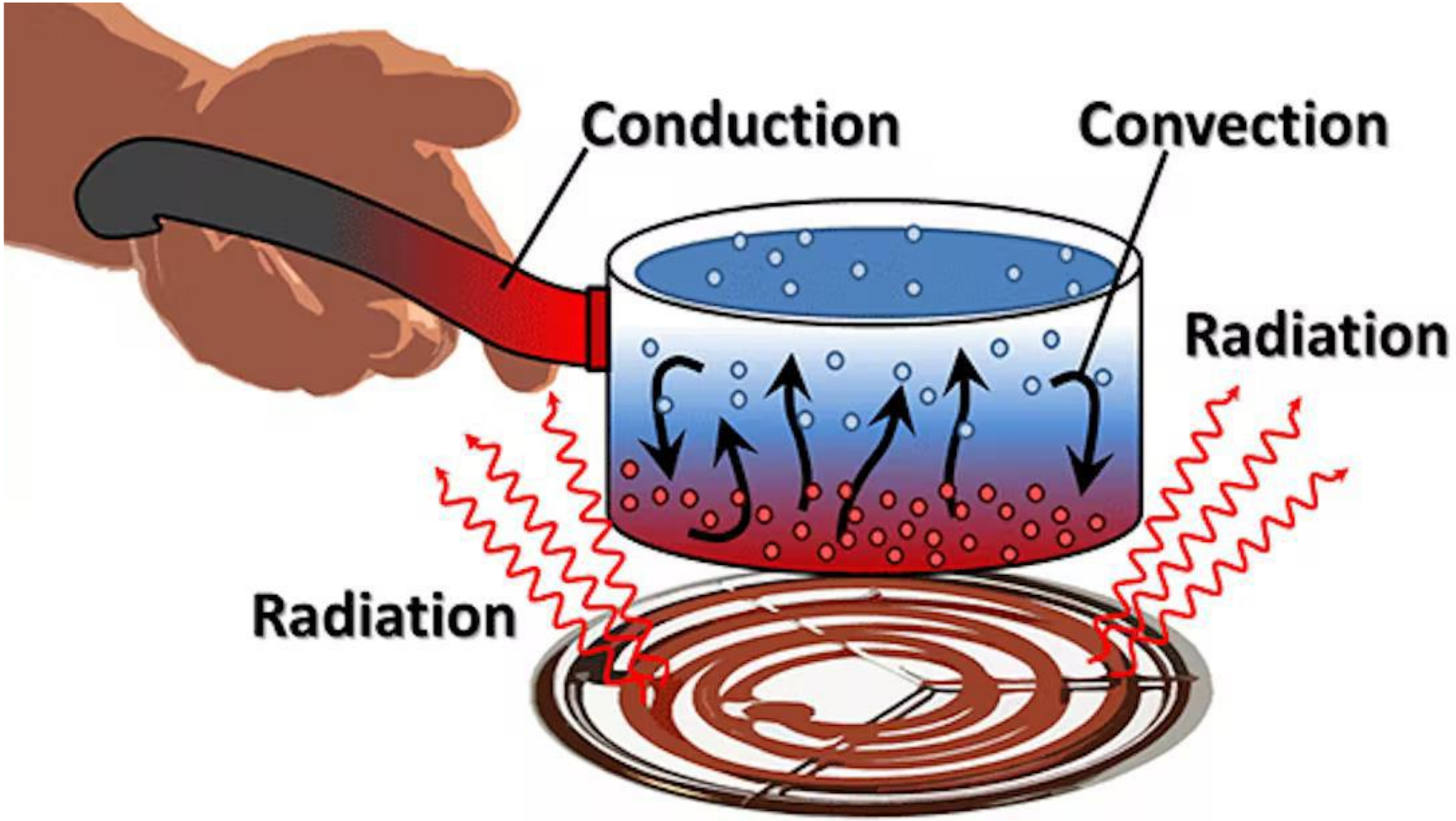
The transfer of energy as heat is always from the higher-temperature medium to the lower-temperature one, and heat transfer stops when the two mediums reach the same temperature.

Heat transfer modes:

- Conduction
- Convection
- Radiation

All modes of heat transfer require the existence of a temperature difference.

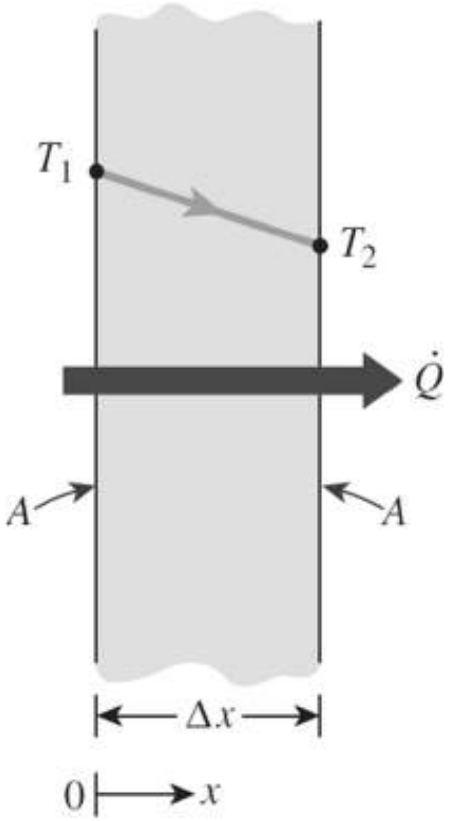
Heat transfer (Heat-flow) types



Conduction Heat Transfer

Conduction: The transfer of energy from the more energetic particles of a substance to the adjacent less energetic ones as a result of interactions between the particles.

- In gases and liquids: Conduction is due to the collisions and diffusion of the molecules during their random motion.
- In solids: It is due to the combination of vibrations of the molecules in a lattice and the energy transport by free electrons.



$$\text{Rate of heat conduction} \propto \frac{(\text{Area})(\text{Temperature difference})}{\text{Thickness}}$$

$$\dot{Q}_{\text{cond}} = kA\frac{T_1 - T_2}{\Delta x} = -kA\frac{\Delta T}{\Delta x} \qquad \text{(kW)}$$

Conduction Heat Transfer

Thermal conductivity (k): A measure of the ability of a material to conduct heat.

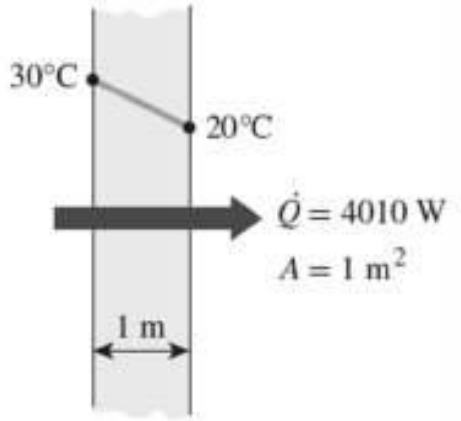
Temperature gradient (dT/dx): The slope of the temperature curve on a T-x diagram.

- Heat is conducted in the direction of decreasing temperature, and the temperature gradient becomes negative when temperature decreases with increasing x.

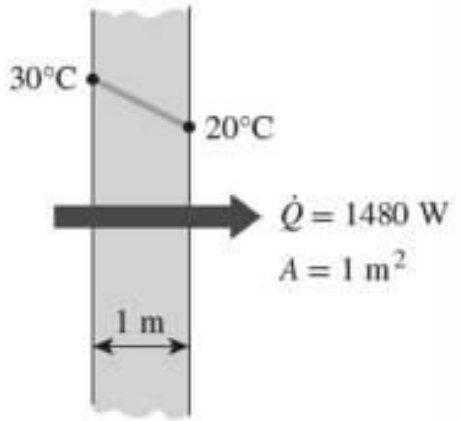
- The negative sign in the equation ensures that heat transfer in the positive x direction is a positive quantity.

$$\dot{Q}_{cond} = -kA\frac{dT}{dx}$$

Fourier's law of Heat conduction



(*a*) Copper ($k = 401$ W/m·K)

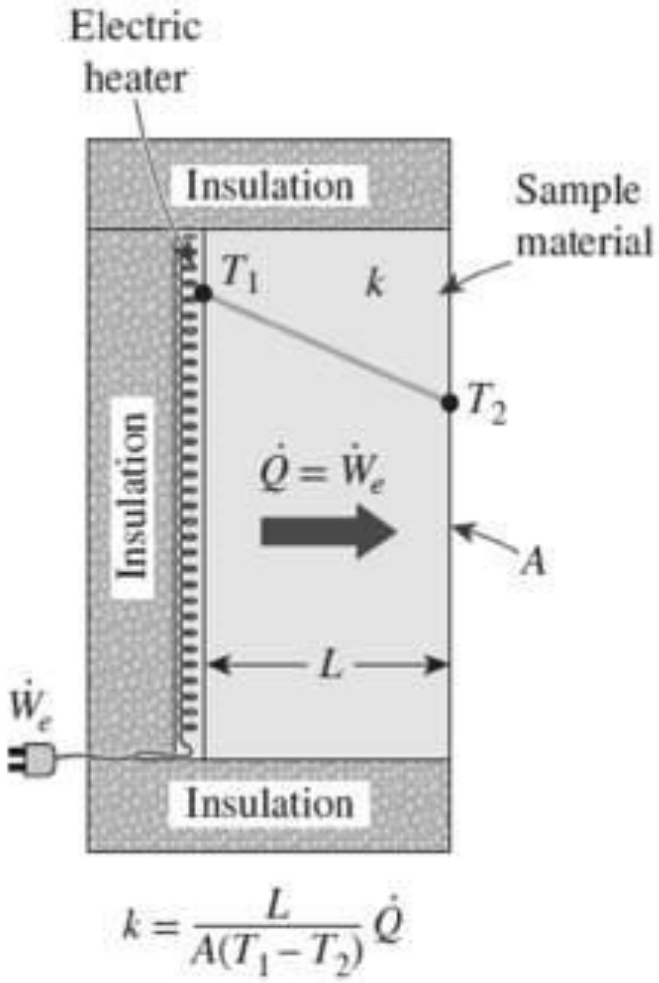


(*b*) Silicon ($k = 148$ W/m·K)

Thermal Conductivity

Thermal conductivity: The rate of heat transfer through a unit thickness of the material per unit area per unit temperature difference.

- The thermal conductivity of a material is a measure of the ability of the material to conduct heat.
- A high value for thermal conductivity indicates that the material is a good heat conductor, and a low value indicates that the material is a poor heat conductor or insulator.



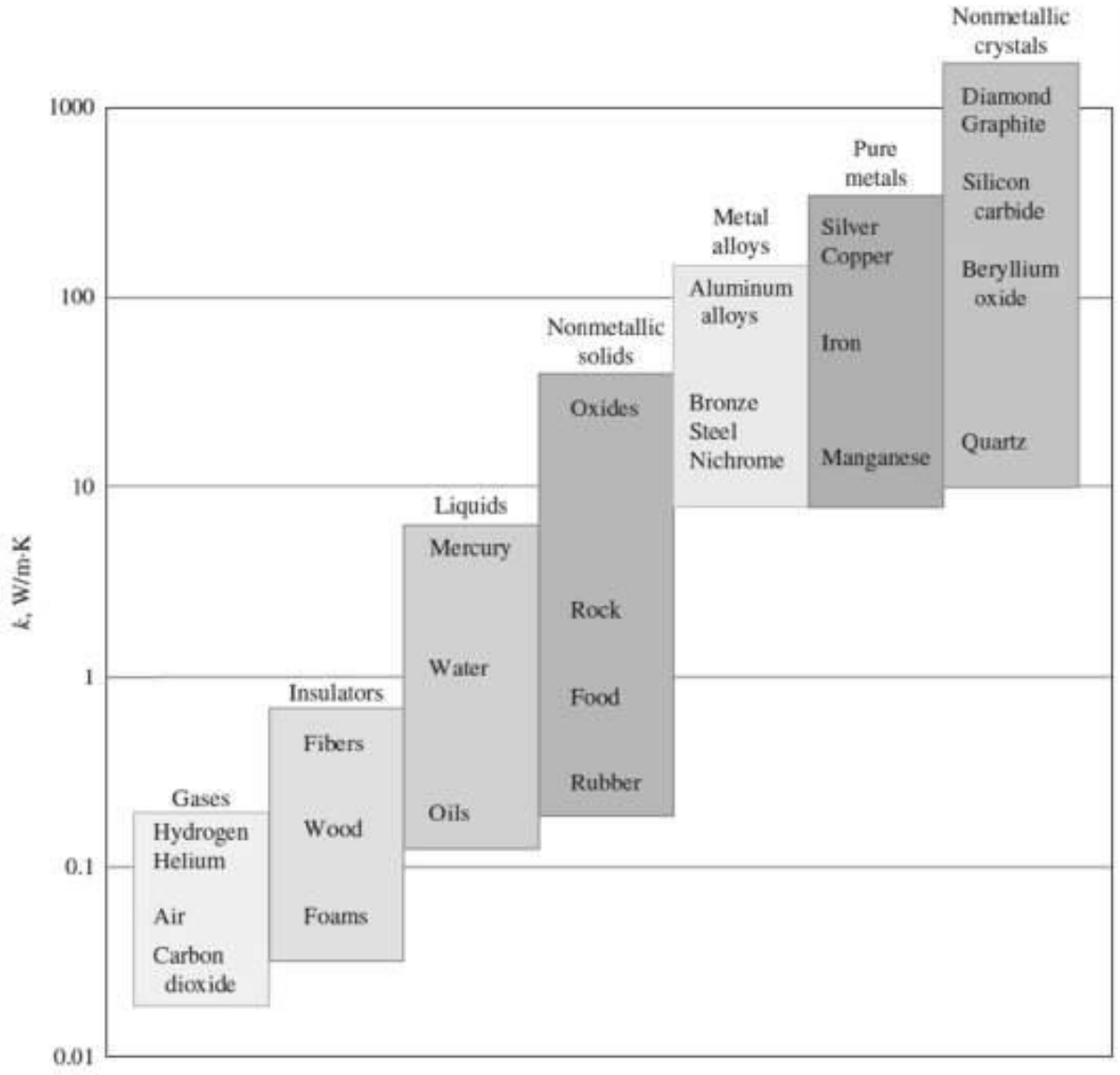
A simple experimental setup to determine the thermal conductivity of a material.

Thermal conductivities of some materials at room temperature

Material	Thermal Conductivity k, W/m·K*
Diamond	2300
Silver	429
Copper	401
Gold	317
Aluminum	237
Iron	80.2
Mercury (l)	8.54
Glass	0.78
Brick	0.72
Water (l)	0.607
Human skin	0.37
Wood (oak)	0.17
Helium (g)	0.152
Soft rubber	0.13
Glass fiber	0.043
Air (g)	0.026
Urethane, rigid foam	0.026

*Multiply by 0.5778 to convert to Btu/h·ft·°F.

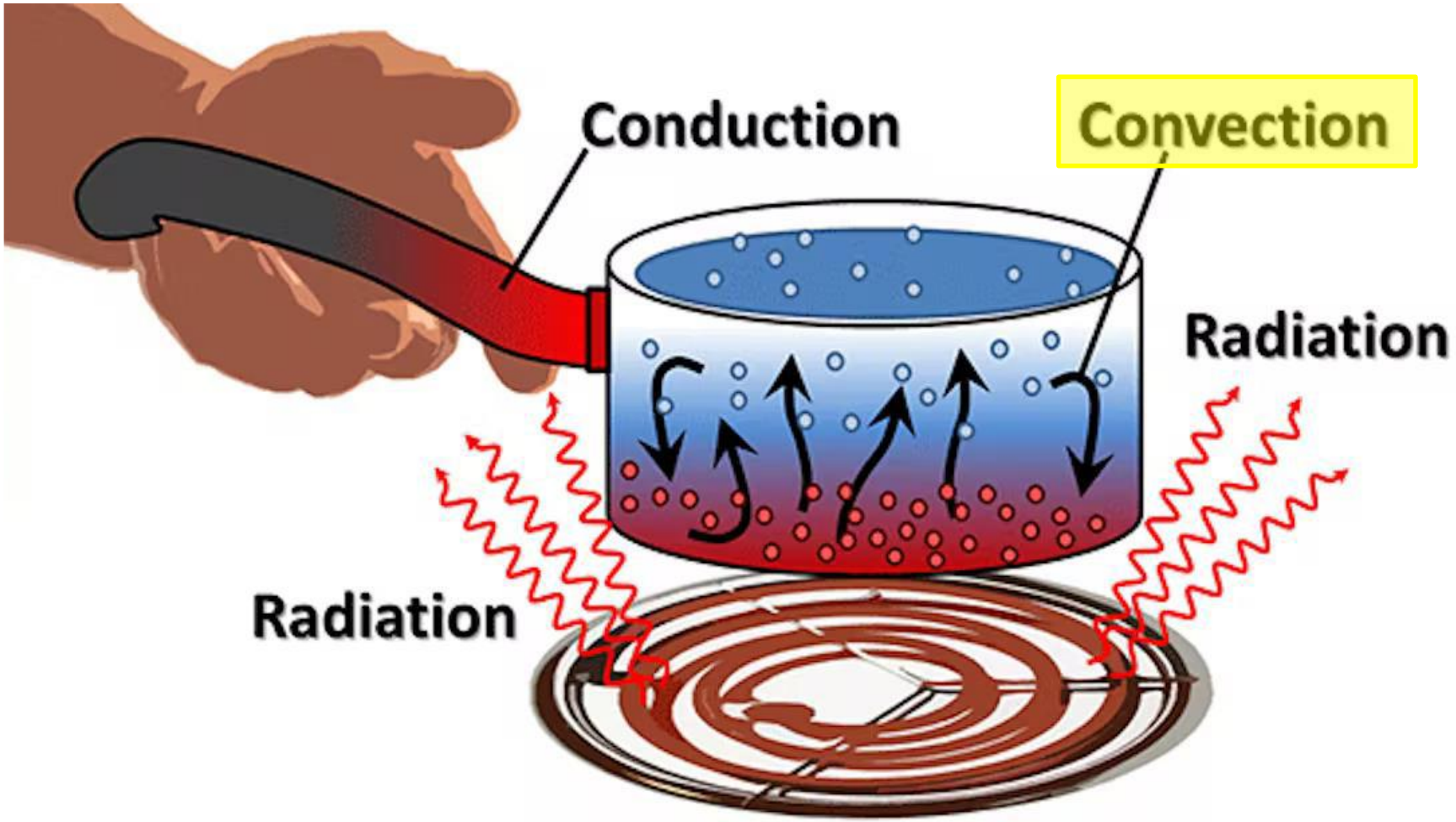
Thermal conductivities of some materials



Thermal conductivities of materials vary with temperature

	k, W/m·K	
T, K	Copper	Aluminum
100	482	302
200	413	237
300	401	237
400	393	240
600	379	231
800	366	218

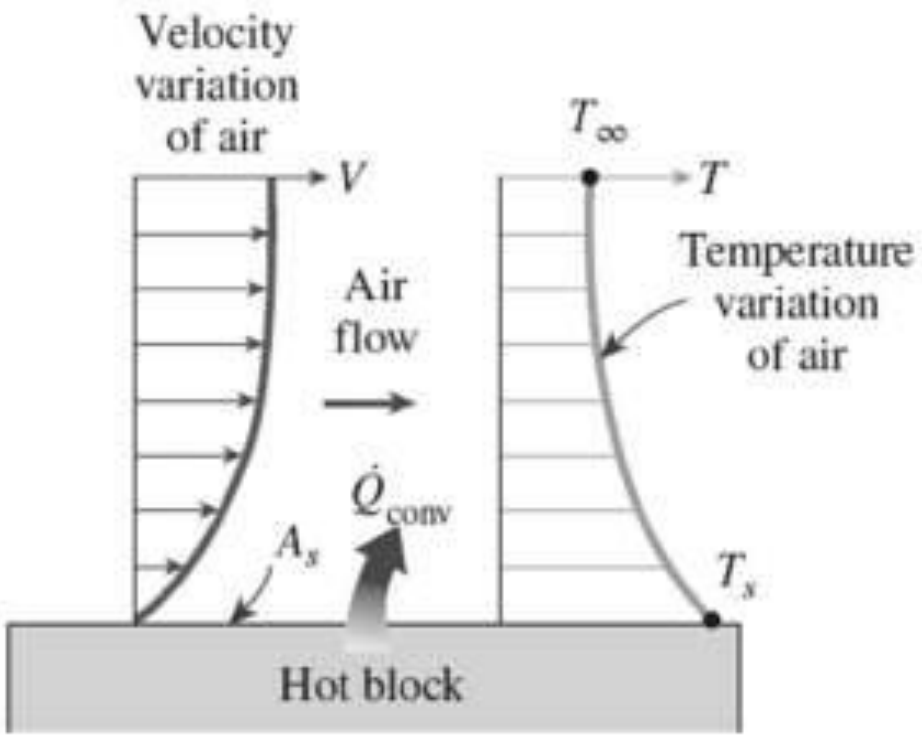
Heat transfer (Heat-flow) types



Convection Heat Transfer

Convection: The mode of energy transfer between a solid surface and the adjacent liquid or gas that is in motion, and it involves the combined effects of conduction and fluid motion.

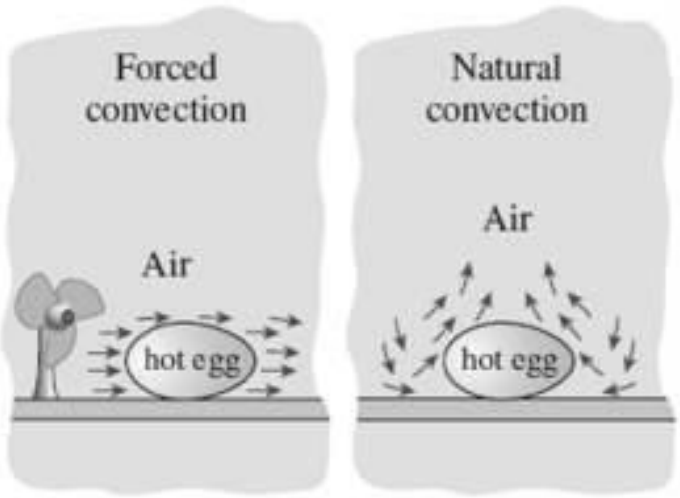
- The faster the fluid motion, the greater the convection heat transfer.
- In the absence of any bulk fluid motion, heat transfer between a solid surface and the adjacent fluid is by pure conduction.



Heat transfer from a hot surface to air by convection

Forced convection: If the fluid is forced to flow over the surface by external means such as a fan, pump, or the wind.

Natural (or free) convection: If the fluid motion is caused by buoyancy forces that are induced by density differences due to the variation of temperature in the fluid.



Heat transfer processes that involve change of phase of a fluid are also considered to be convection because of the fluid motion induced during the process, such as the rise of the vapor bubbles during boiling or the fall of the liquid droplets during condensation.

$$\dot{Q}_{conv} = hA_s(T_s - T_\infty) \quad (kW)$$

Newton's law of cooling

h	convection heat transfer coefficient, $W/m^2 \cdot °C$
A_s	the surface area through which convection heat transfer takes place
T_s	the surface temperature
T_∞	the temperature of the fluid sufficiently far from the surface.

Convection heat transfer coefficient h:

Not a property of the fluid. It is an experimentally determined parameter.

h depends on:

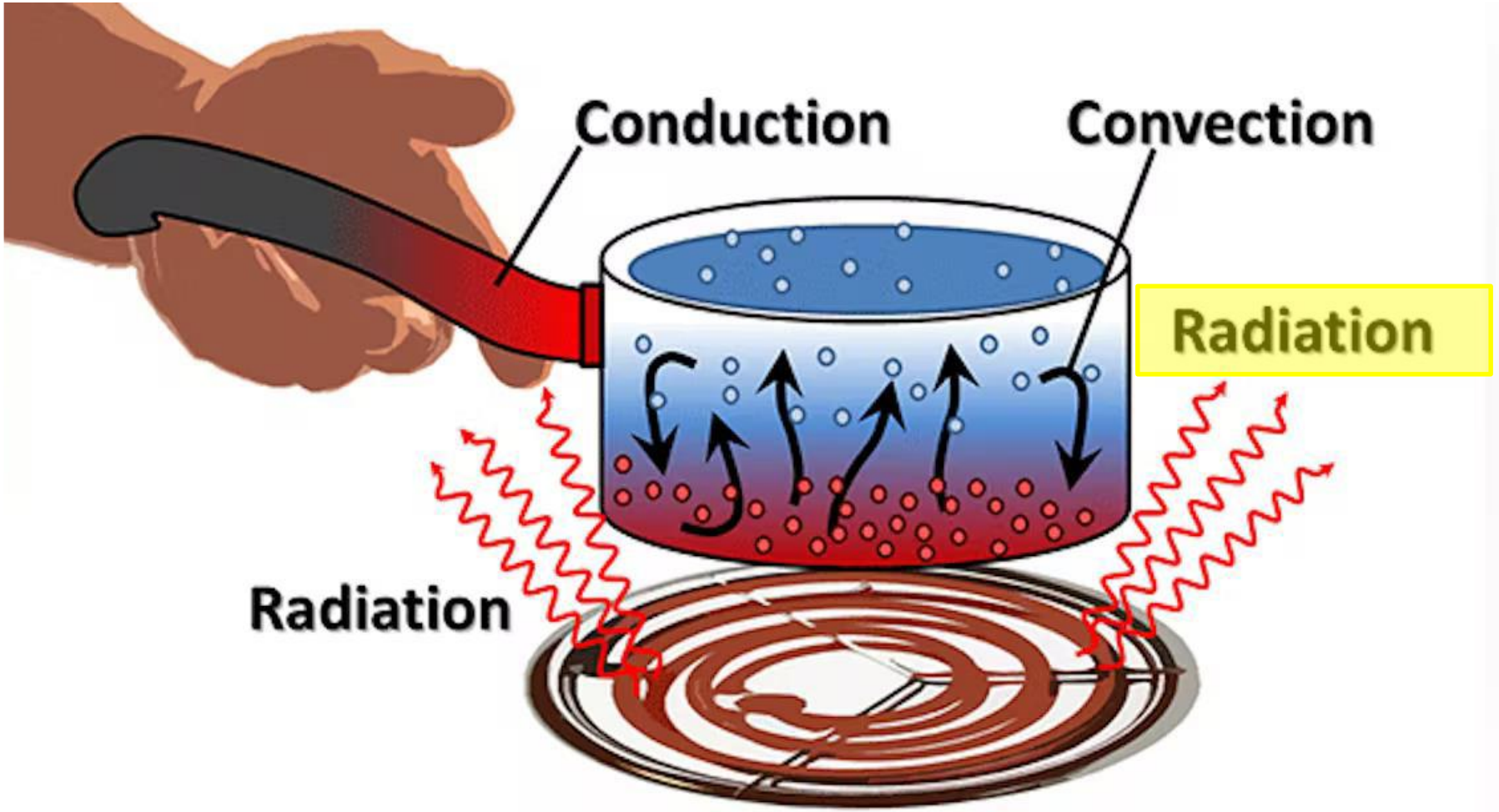
- the surface geometry
- the nature of fluid motion
- the properties of the fluid
- the bulk fluid velocity

Typical values of Convection Heat Transfer Coefficient

Type of Convection	Heat Transfer Coefficient h, $W/m^2 \cdot K^*$
Free convection of gases	2–25
Free convection of liquids	10–1000
Forced convection of gases	25–250
Forced convection of liquids	50–20,000
Boiling and condensation	2500–100,000

*Multiply by 0.176 to convert to $Btu/h \cdot ft^2 \cdot °F$.

Heat transfer (Heat-flow) types



Radiation Heat Transfer

Radiation: The energy emitted by matter in the form of electromagnetic waves (or photons) as a result of the changes in the electronic configurations of the atoms or molecules.

Unlike conduction and convection, the transfer of heat by radiation does not require the presence of an intervening medium.

In fact, heat transfer by radiation is fastest (at the speed of light) and it suffers no attenuation in a vacuum.

This is how the energy of the sun reaches the earth.

Thermal radiation: In heat transfer studies we are interested in thermal radiation, which is the form of radiation emitted by bodies because of their temperature.

All bodies at a temperature above absolute zero emit thermal radiation.

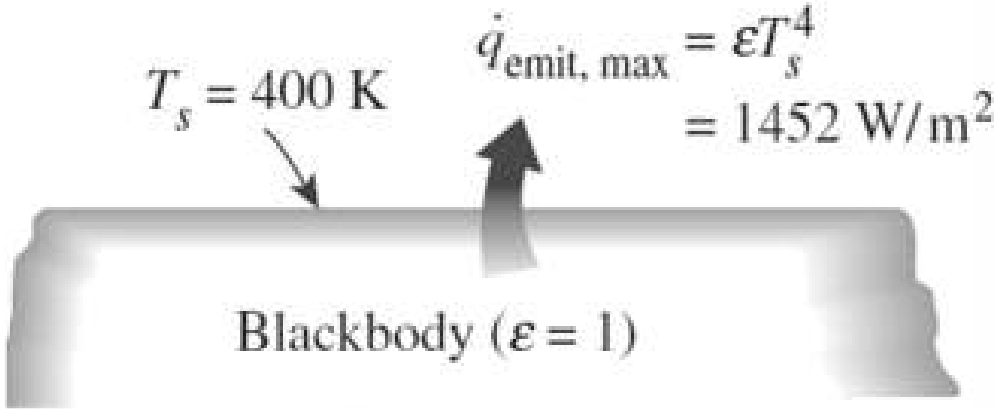
$$\dot{Q}_{\text{emit,max}} = \sigma A_s T_s^4 \qquad \text{(kW)}$$ **Stefan–Boltzmann law**

$\sigma = 5.670 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4$ Stefan–Boltzmann constant

Blackbody: The idealized surface that emits radiation at the maximum rate.
Radiation emitted by real surfaces

$$\dot{Q}_{\text{emit}} = \varepsilon \sigma A_s T_s^4 \qquad \text{(kW)}$$

Emissivity ε : A measure of how closely a surface approximates a blackbody for which $\varepsilon = 1$ of the surface. $0 \le \varepsilon \le 1$



Emissivities of some materials at 300 K

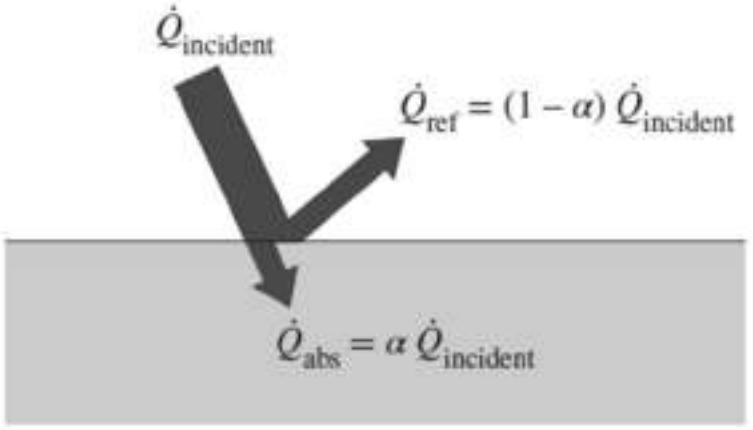
Material	Emissivity ε
Aluminum foil	0.07
Anodized aluminum	0.82
Polished copper	0.03
Polished gold	0.03
Polished silver	0.02
Polished stainless steel	0.17
Black paint	0.98
White paint	0.90
White paper	0.92–0.97
Asphalt pavement	0.85–0.93
Red brick	0.93–0.96
Human skin	0.95
Wood	0.82–0.92
Soil	0.93–0.96
Water	0.96
Vegetation	0.92–0.96

Absorptivity α: The fraction of the radiation energy incident on a surface that is absorbed by the surface. $0 \leq \alpha \leq 1$

A blackbody absorbs the entire radiation incident on it ($\alpha = 1$).

Kirchhoff's law: The emissivity and the absorptivity of a surface at a given temperature and wavelength are equal.

$$\dot{Q}_{\text{absorbed}} = \alpha \dot{Q}_{\text{incident}} \qquad \text{(kW)}$$



The absorption of radiation incident on an opaque surface of absorptivity.

Net radiation heat transfer: The difference between the rates of radiation emitted by the surface and the radiation absorbed.

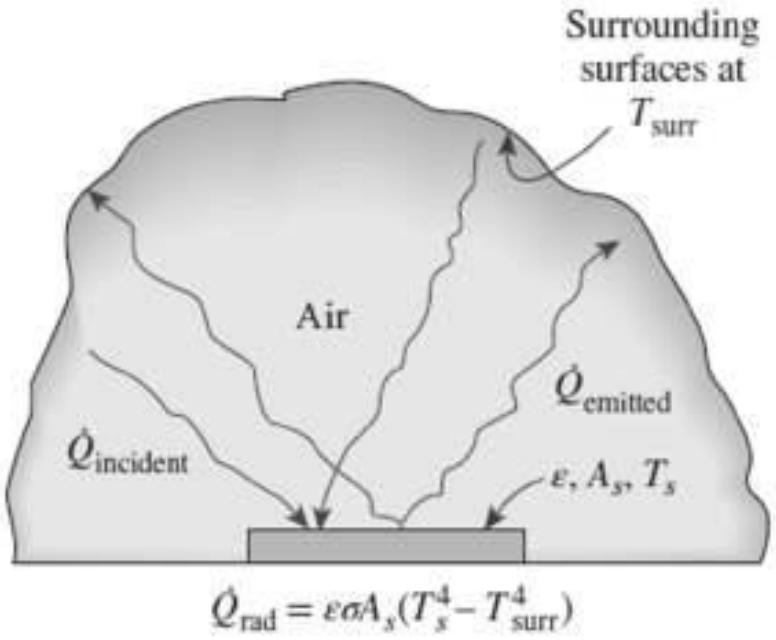
The determination of the net rate of heat transfer by radiation between two surfaces is a complicated matter since it depends on

- the properties of the surfaces
- their orientation relative to each other
- the interaction of the medium between the surfaces with radiation

Radiation is usually significant relative to conduction or natural convection, but negligible relative to forced convection.

When a surface is completely enclosed by a much larger (or black) surface at temperature Tsurr separated by a gas (such as air) that does not intervene with radiation, the net rate of radiation heat transfer between these two surfaces is given by

$$\dot{Q}_{\text{rad}} = \varepsilon \sigma A_s (T_s^4 - T_{\text{surr}}^4) \qquad \text{(kW)}$$



Radiation heat transfer between a surface and the surfaces surrounding it.

When radiation and convection occur simultaneously between a surface and a gas:

$$\dot{Q}_{\text{total}} = h_{\text{combined}} A_s (T_s - T_\infty) \qquad \text{(W)}$$

Combined heat transfer coefficient $h_{combined}$ includes the effects of both convection and radiation

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = h_{\text{conv}} A_s (T_s - T_\infty) + \varepsilon \sigma A_s (T_s^4 - T_{\text{surr}}^4)$$

$$\dot{Q}_{\text{total}} = h_{\text{combined}} A_s (T_s - T_\infty)$$

$$h_{\text{combined}} = h_{\text{conv}} + h_{\text{rad}} = h_{\text{conv}} + \varepsilon \sigma (T_s + T_{\text{surr}})(T_s^2 + T_{\text{surr}}^2)$$


THANK YOU